



NORTH SYDNEY BOYS HIGH SCHOOL

2012 HSC ASSESSMENT TASK 2

Mathematics

General Instructions

Section I contains 4 multiple choice questions to be answered on the separate answer sheet provided.

Reading Time – 5 minutes
Working Time – 55 minutes
Total Time – 60 minutes

Section II is to be answered in the booklet provided, showing ALL necessary working.

- Write using blue or black pen
- Board approved calculators may be used
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher: Please tick

- ☐ Mr Berry
- ☐ Ms Ziariaris
- ☐ Mr Lowe
- ☐ Mr Weiss
- ☐ Mr Lam
- ☐ Mr Ireland
- ☐ Mr Fletcher

Student Number: _____

(To be used by the exam markers only.)

Question No	1-4	5	6	7	8	9	Total	Total %
Mark	4	9	11	9	4	6	43	100

SECTION I

Question 1

The parabola whose equation is $x^2 = \frac{y}{4}$ has focal length equal to

1

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 4 (D) $\frac{1}{16}$

Question 2

The parabola with equation $(y-3)^2 = 16(x+4)$ has its vertex at

1

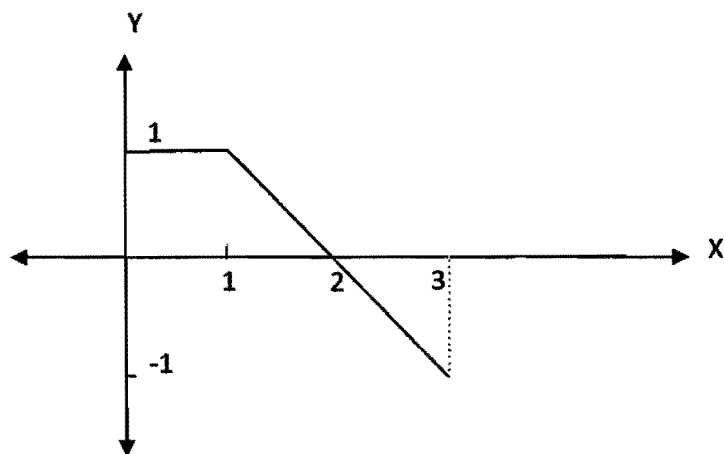
- (A) (3, -4) (B) (-3, 4) (C) (4, -3) (D) (-4, 3)

Question 3

Given the graph of $y = f(x)$ below, the value of $\int_0^3 f(x) dx$ is

1

- (A) 0 (B) 1 (C) 2 (D) 3



Question 4

$$\int \frac{dx}{2} =$$

1

- (A) $2x + C$ (B) $2^0 + C$ (C) $\frac{1}{2}x + C$ (D) $-\frac{1}{2}x + C$

SECTION II

Question 5

Find the indefinite integrals

- a) $\int (6x - x^4) dx$ 1
- b) $\int (3x - 5)^4 dx$ 1
- c) $\int (2x^2 - 1)^2 dx$ 2

Find the definite integrals

- d) $\int_0^1 \frac{dx}{(8x+5)^2}$ 2
- e) $\int_4^9 \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx$ 3

Question 6

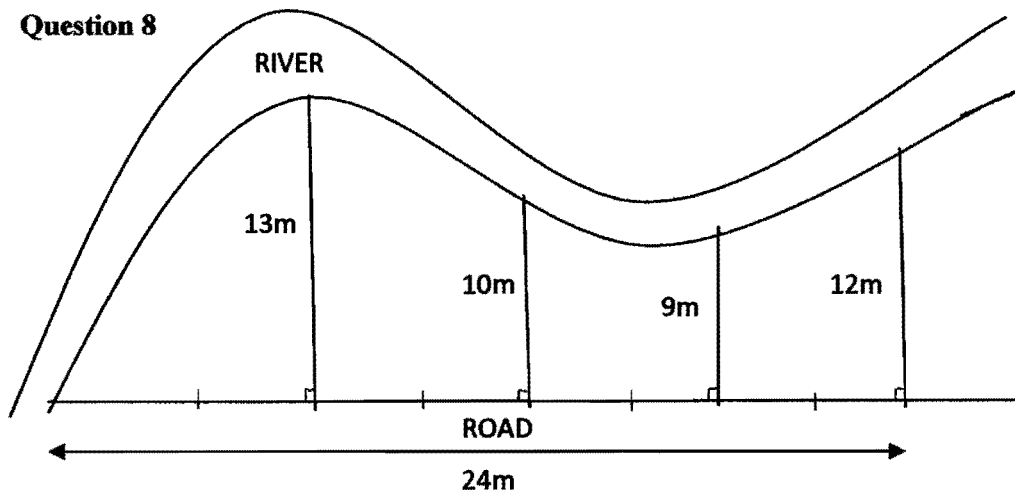
- a) Write down the equation of the parabola with focus $(0, -2)$ and directrix $y = 2$. 1
- b) Write down the equation of the parabola with focus $(3, 0)$ and vertex $(-1, 0)$. 1
- c) Find the equation of the parabola with vertex $(-2, 3)$, axis parallel to the Y-axis and passing through the point $(-6, -5)$. 3
- d) For the equation $x = 4y - y^2$, find
- i. the vertex 3
 - ii. the focal length 1
 - iii. the focus 1
 - iv. the equation of the directrix 1

Question 7

- a) Calculate the area under the curve $y = x^2 + 1$ bounded by the X-axis and $x = -1$ and $x = 1$. 2
- b) If the gradient function is $\frac{dy}{dx} = \frac{3x^4 - 1}{x^2}$ and $y = 10$ when $x = 2$, find the primitive function. 3
- c) Find the area between $y = (x - 2)(x - 1)$ and $y = x - 1$. 4

Question 8

a)

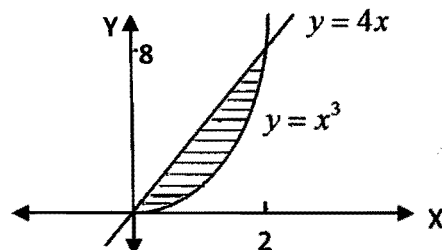


Wasteland bordering a river bank and a straight road was fenced off and used as a recreational park. Perpendicular distances from the road to the river bank are shown on the diagram.

- Use the Simpson's Rule to approximate the area of the recreational park. 2
- Use the Trapezoidal Rule to approximate the area of the recreational park. 2

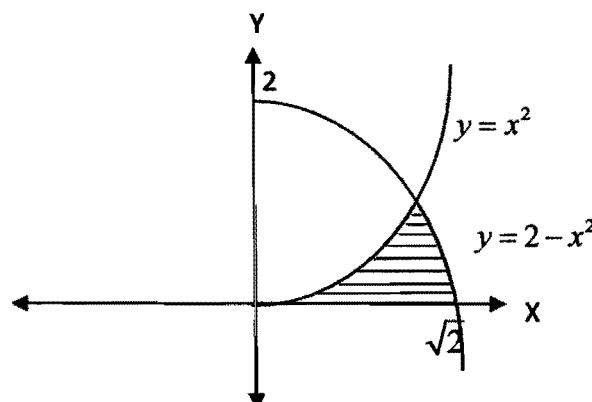
Question 9

- a) The shaded region between the curves $y = x^3$ and $y = 4x$ in the first quadrant in the graph below is rotated about the Y-axis. Write down the integral that would calculate the volume of the solid formed. **DO NOT EVALUATE THE INTEGRAL.**



2

- b) The shaded area in the diagram below enclosed by $y = x^2$, $y = 2 - x^2$ and the X-axis is rotated about the X-axis. Find the volume of the solid generated correct to 2 decimal places.



4

1. $4a = \frac{1}{4}$ D.
 $a = \frac{1}{16}$

2. $(-4, 3)$ D.

3. $1 + \frac{1}{2} - \frac{1}{2}$ B
 $= 1$

4. C.

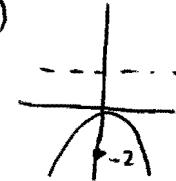
5. a) $\frac{6x^2}{2} - \frac{x^5}{5} + C$
 $3x^2 - \frac{x^5}{5} + C$

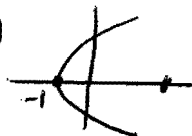
b) $\frac{(3x-5)^5}{15} + C$

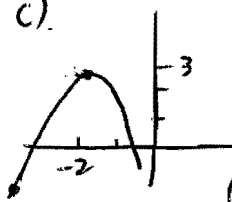
c) $\int (4x^4 - 4x^2 + 1) dx$
 $= \frac{4x^5}{5} - \frac{4x^3}{3} + x + C$

d) $\int_0^1 (8x+5)^{-2} dx$
 $= \left[\frac{(8x+5)^{-1}}{-8} \right]_0^1$
 $= \frac{1}{8} \left[\frac{-1}{8x+5} \right]_0^1$
 $= \frac{1}{8} \left[-\frac{1}{13} + \frac{1}{5} \right]$
 $= \frac{1 \times 8}{8 \times 65}$
 $= \frac{1}{65}$

e) $\int_4^9 \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx$
 $= \int_4^9 (x^{\frac{1}{2}} + x^{-\frac{1}{2}}) dx$
 $= \left[\frac{2x^{\frac{3}{2}}}{3} + 2x^{\frac{1}{2}} \right]_4^9$
 $= \left[\frac{2}{3} \times 9^{\frac{3}{2}} + 2 \times 9^{\frac{1}{2}} - \left(\frac{2}{3} \times 4^{\frac{3}{2}} + 2 \times 4^{\frac{1}{2}} \right) \right]$
 $= 24 - 9\frac{1}{3}$
 $= 14\frac{2}{3}$

b.a)  $x^2 = -8y$

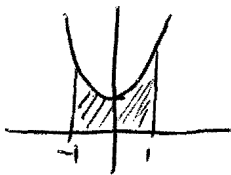
b)  $y^2 = 16(x+1)$

c)  $(x+2)^2 = -4a(y-3)$
Sub $(-6, -5)$
 $(-6+2)^2 = -4a(-8)$
 $16 = 32a$
 $a = \frac{1}{2}$
 $\therefore \text{Eqn: } (x+2)^2 = -2(y-3)$

d) (i) $y^2 - 4y = -x$
 $y^2 - 4y + (-2)^2 = -x + 4$
 $(y-2)^2 = -(x-4)$
Vertex $(4, 2)$

(ii) $4a = 1$ (iii) Focus $(3\frac{3}{4}, 2)$
 $a = \frac{1}{4}$ (iv) $x = 4\frac{1}{4}$

7.a)



$$\begin{aligned}
 A &= 2 \int_0^1 (x^2 + 1) dx \\
 &= 2 \left[\frac{x^3}{3} + x \right]_0^1 \\
 &= 2 \left[\frac{1}{3} + 1 \right] \\
 &= \frac{8}{3} \text{ sq. units.}
 \end{aligned}$$

b) $\frac{dy}{dx} = \frac{3x^4 - 1}{x^2}$

$$= 3x^2 - x^{-2}$$

$$y = \int 3x^2 - x^{-2} dx$$

$$y = x^3 - \frac{x^{-1}}{-1} + C$$

$$y = x^3 + \frac{1}{x} + C$$

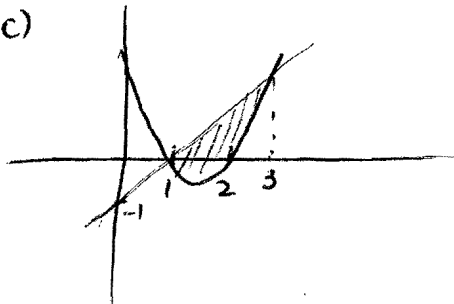
Sub $y=10, x=2$

$$10 = 8 + \frac{1}{2} + C$$

$$C = 1\frac{1}{2}$$

$$\therefore y = x^3 + \frac{1}{x} + \frac{3}{2}$$

c)



Points of Intersection

$$(x-2)(x-1) = x-1$$

$$(x-1)(x-2-1) = 0$$

$$x=1, x=3$$

$$y=0 \quad y=2$$

$$\begin{aligned}
 A &= \int_1^3 (x-1) - (x-2)(x-1) dx \\
 &= \int_1^3 x-1 - (x^2-3x+2) dx \\
 &= \int_1^3 -x^2+4x-3 dx \\
 &= \left[-\frac{x^3}{3} + 2x^2 - 3x \right]_1^3 \\
 &= \left[-9 + 18 - 9 - \left(-\frac{1}{3} + 2 - 3 \right) \right] \\
 &= \left[\frac{4}{3} \right] \text{ sq. units.}
 \end{aligned}$$

8.a) (i)

$$\begin{aligned}
 A &\div \frac{6}{3} \{ 0 + 4(13+9) + 2(10) + 12 \} \text{ ①} \\
 &= 240 \text{ sq. units.} \quad \text{①}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad A &\div \frac{6}{2} \{ 0 + 2(13+10+9) + 12 \} \text{ ①} \\
 &= 228 \text{ sq. units.} \quad \text{①}
 \end{aligned}$$

$$\begin{aligned}
 9.a) \quad V &= \pi \int_0^8 \frac{y^2}{3} dy - \pi \int_0^8 \frac{y^2}{16} dy \text{ ①} \\
 &= \pi \left[\frac{3y^{5/3}}{5} \right]_0^8 - \pi \left[\frac{y^3}{48} \right]_0^8
 \end{aligned}$$

b) Point of Intersection

$$x^2 = 2 - x^2$$

$$2x^2 = 2$$

$$x^2 = 1$$

$$x = \pm 1$$

$$\begin{aligned}
 V &= \pi \int_0^1 x^4 dx + \pi \int_1^{\sqrt{2}} (4 - 4x^2 + x^4) dx \text{ ①} \\
 &= \pi \left[\frac{x^5}{5} \right]_0^1 + \pi \left[4x - \frac{4x^3}{3} + \frac{x^5}{5} \right]_1^{\sqrt{2}} \text{ ①} \\
 &= \pi \left(\frac{1}{5} \right) + \pi \left(4\sqrt{2} - \frac{8\sqrt{2}}{3} + \frac{4\sqrt{2}}{5} - \left(4 - \frac{4}{3} + \frac{1}{5} \right) \right) \\
 &= 1.10u^3 \text{ ①} \quad (1.35\pi)
 \end{aligned}$$